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3-3: The Square with a Missing Corner

TEKS P.5.J: solve polynomial equations by applying a variety of techniques. Today's technique: completing the square.

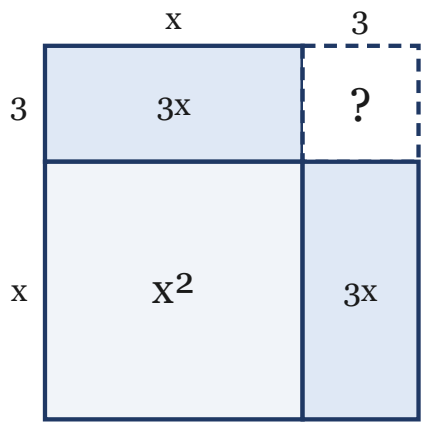
Warm-up: Squares and square roots

(a) Expand: $(x + 3)^2$ = _____ **(b)** Expand: $(x - 5)^2$ = _____

(c) Solve exactly: $x^2 = 20$. x = _____

(d) Look at (a) and (b). How is the last number (9, 25) related to the middle coefficient (6, −10)?

Explore: Fill in the missing corner



Not to scale. Shaded = pieces we have; dashed = missing.

The picture shows **$x^2 + 6x$**: an x-by-x square and two strips, each 3 wide. It is almost one big square.

(a) The missing corner is _____ by _____, so its area is _____.

(b) Fill it in: $x^2 + 6x +$ _____

(c) The big square has side $x +$ _____, so $x^2 + 6x + 9 = (x +$ _____$)^2$.

(d) The 3 is what fraction of the 6? The 9 is 3 _____.

Now use it. Suppose **$x^2 + 6x = 40$**. Fill in each blank (remember the equation must stay balanced):

$x^2 + 6x +$ _____ $= 40 +$ _____ $\rightarrow (x +$ _____$)^2 =$ _____ $\rightarrow x + 3 = \pm$ _____ $\rightarrow x =$ _____ or $x =$ _____

1. B Find the number that completes the square, then write the perfect square.

(a) $x^2 + 10x +$ _____ $= (x +$ _____$)^2$ (b) $x^2 - 8x +$ _____ $= (x -$ _____$)^2$ (c) $x^2 + 7x +$ _____ $= (x +$ _____$)^2$

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Practice

B = Basic level (leading coefficient 1). Give **exact** answers (simplified radicals and fractions). Show your steps. Check at least one solution by substituting.

2. **B** Solve: $(x + 3)^2 = 16$

3. **B** Solve: $(x - 5)^2 = 7$

4. **B** Solve by completing the square: $x^2 + 8x + 7 = 0$

5. **B** Solve by completing the square: $x^2 - 8x + 5 = 0$

6. **B** (a) Rewrite $x^2 - 6x - 4 = 0$ in the form $(x - p)^2 = q$. $p =$ _____ $q =$ _____ (b) Solve it. (c) Check $x = 3 + \sqrt{13}$ in the **original** equation. (Hint: $(3 + \sqrt{13})^2 = 22 + 6\sqrt{13}$.)

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7. B Solve: $x^2 + 5x - 2 = 0$ (b is odd: expect fractions.)

8. B Solve: $x^2 - 3x - 10 = 0$

9. B **Preview.** Solve: $x^2 + 4x + 13 = 0$. (You know i from Algebra II. What happens when the right side is negative?)

10. B Solve: $x^2 = 2x + 9$ (Move terms first.)

11. B A rectangular label is **6 cm longer** than it is wide. Its area is **7 cm²**. Let x = the width in cm.

(a) Complete: x(_____) = 7, so x^2 + _____x = 7.

(b) Which solution makes sense? Give the dimensions and the perimeter.

_____

Math Time Machine: Around the year 820, al-Khwarizmi wrote a book whose title includes *al-jabr* ("completion"), and his method for quadratics was, in effect, completing a square drawn with a missing corner. Babylonian scribes, thousands of years earlier, used a procedure that matches it, with no symbols at all.